DATABASE MANAGEMENT SYSTEM (ENCS205)

Lab Report Submitted to

K.R. Mangalam University

for

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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Experiment No:1

Aim:

To design an Entity Relationship (ER) diagram for the College database and convert the ER diagram into relational tables.

Description:

The College database maintains information about student , semester & section , subjects, class allotment, and internal assessment marks.Each student belongs to a particular semester and section and is evaluated for different subjects.

Entities and Attributes:

1.STUDENT

- USN (Primary Key)
- SName
- Address
- Phone

Gender

2. SEMSEC

- SSID (Primary Key)
- Sem

Sec

3. SUBJECT

- Subcode (Primary Key)
- Title

Sem
Credits

5. IAMARKS

- USN (Foreign Key)
- Subcode (Foreign Key)
- SSID (Foreign Key)
- Test1

• Test2
• Test3
Final IA

Relationships:

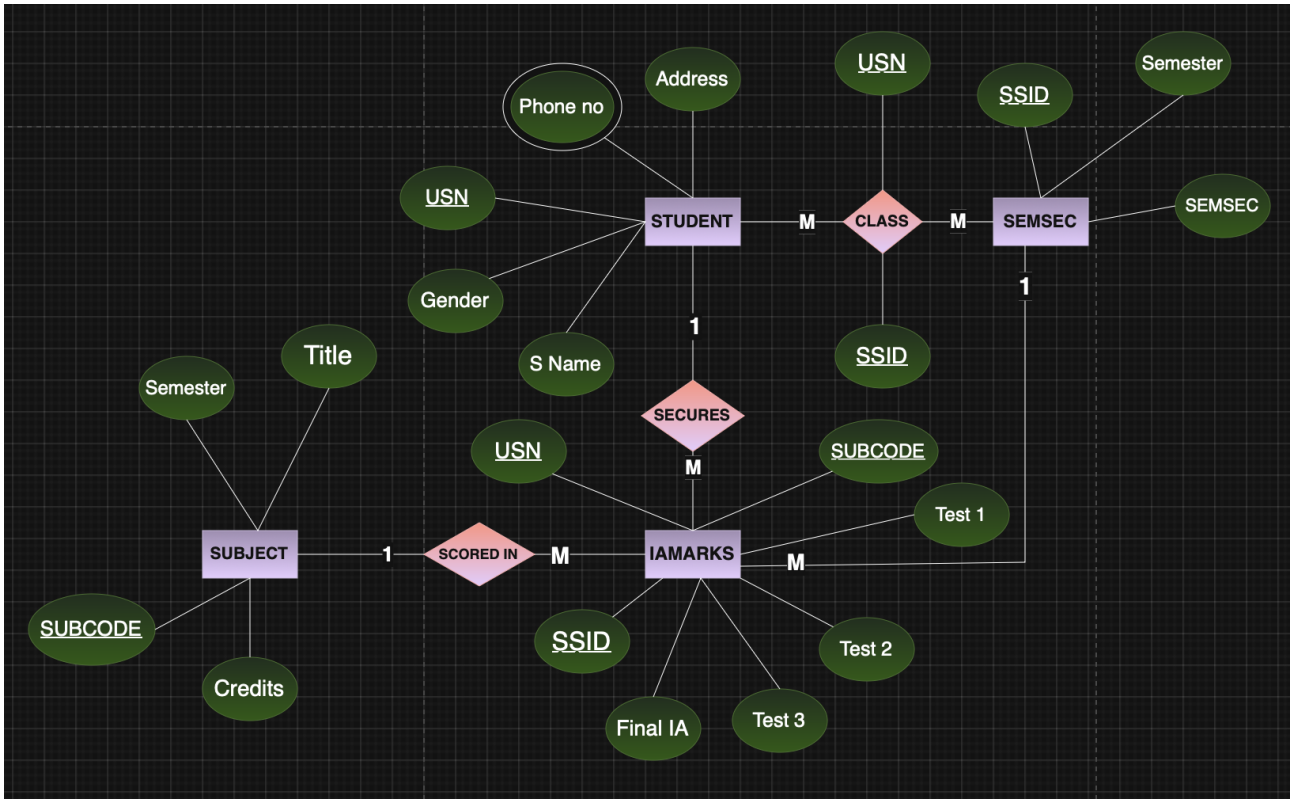
• **CLASS** relates **STUDENT** and **SEMSEC** in an M:N relationship.

SECURES relates **STUDENT** and **IAMARKS** in a 1:M relationship.

SCORED IN relates **SUBJECT** and **IAMARKS** in a 1:M relationship.

SEMSEC has a 1:M relationship with **IAMARKS.**

ER diagram:



Conclusion:

The ER diagram represents a well-structured database design with properly defined entities, attributes, and relationships, ensuring data integrity, minimal redundancy, and efficient data organization for managing student internal assessment records.

Experiment No: 2

Aim:

To design an Entity Relationship (ER) diagram for the Company database and convert it into relational tables.

Description:

The Company database stores information about employees, departments,
projects, locations, and employee participation in projects.Employees work in departments and may work on multiple projects.

Entities and Attributes:

1. EMPLOYEE

• SSN (Primary Key)
• Name
• Address
• Sex
• Salary
• SuperSSN
DNo

2. DEPARTMENT

• DNo (Primary Key)
DName
Mgr SSN
Mgr StartDate

3. DLOCATION

• DNo (Foreign Key)
DLoc

4. PROJECT

• PNo (Primary Key)

• PName
• PLocation
DNo

5. WORKS_ON

• SSN (Foreign Key)
• PNo (Foreign Key)
Hours

Relationships:

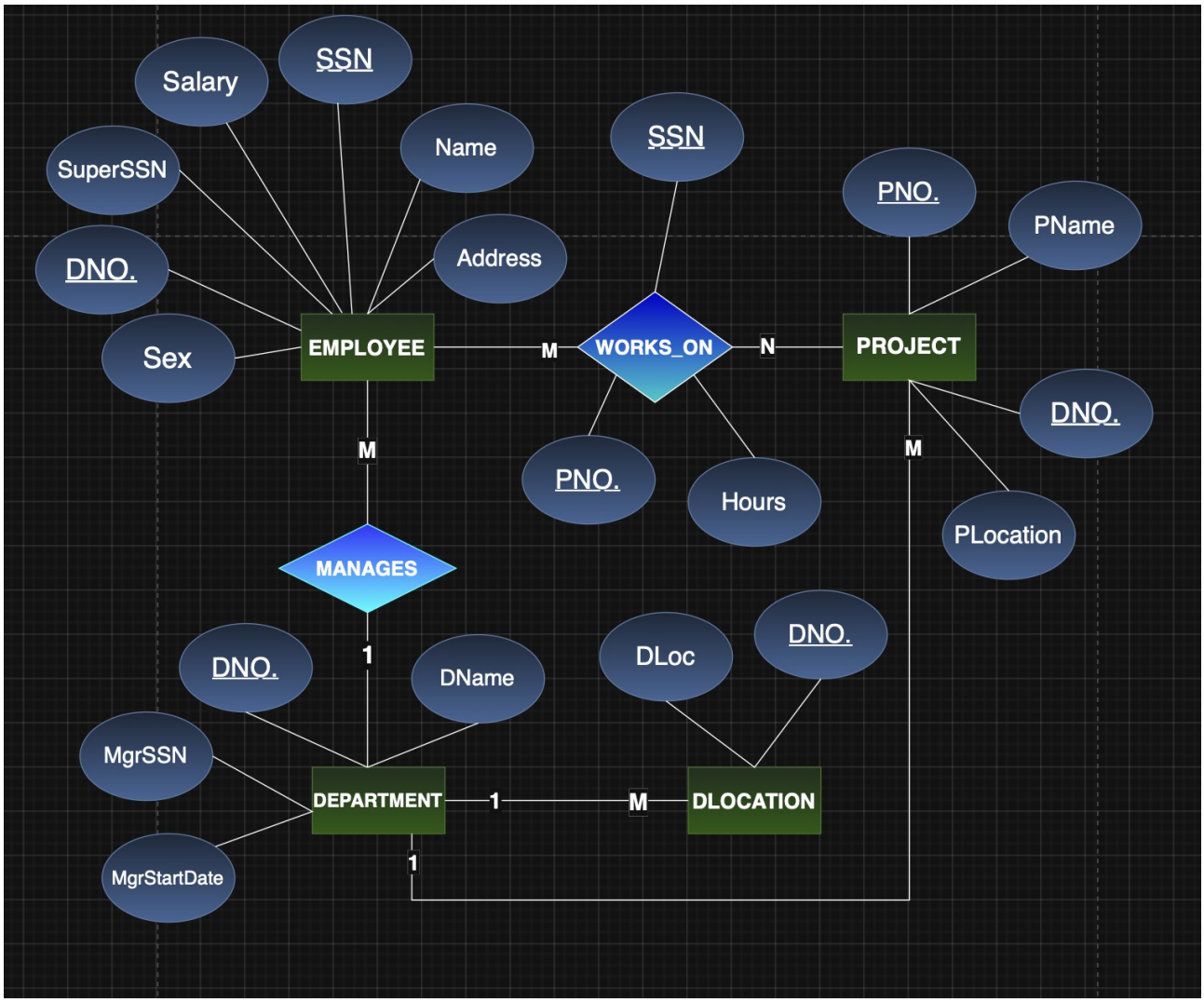
•EMPLOYEE – WORKS_ON – PROJECT
→ M : N relationship **(Hours is a relationship attribute)**

EMPLOYEE – MANAGES – DEPARTMENT→ 1 : 1 relationship

DEPARTMENT – DLOCATION→ 1 : M relationship

DEPARTMENT – EMPLOYEE→ 1 : M relationship

ER diagram:



Conclusion:

The ER diagram represents a normalized organizational database with well-defined entities and relationships, ensuring data consistency and efficient data management.